1. **Project Title**

Loan Default Risk Prediction Using Machine Learning

2. **Problem Statement**

Financial institutions face significant challenges in identifying customers who are likely to default on loans. Incorrect assessment of credit risk can lead to financial losses and poor decision-making in loan approvals. Therefore, there is a need for an intelligent system that can accurately predict loan default risk using available customer data.

This project aims to build a machine learning model that analyzes demographic, financial, and credit-related features to classify whether a customer will default or not. The problem is formulated as a binary classification task, where the output indicates default risk (1) or no default (0).

By predicting high-risk customers in advance, the model can help institutions reduce credit risk, improve loan approval strategies, and enhance overall financial stability. The solution should be accurate, reliable, and scalable for real-world financial applications.

3. **Objective**

The main objective of this project is to predict whether a customer is likely to default on a loan.

This system helps financial institutions to:

Reduce credit risk

Identify high-risk customers early

Make accurate and data-driven loan approval decisions

4. **Dataset Description**

The dataset contains customer demographic and financial information such as:

Income

Savings

Monthly expenses

Loan amount

Credit score

Employment details

Loan history

The target variable is Loan Default Status, where:

0 = No Default

1 = Default

5. **Methodology**

The project follows a complete end-to-end machine learning workflow:

Data loading and inspection

Exploratory Data Analysis (EDA)

Handling missing values using imputation

Feature scaling and encoding

Train-test split

Model training

Model evaluation and comparison

6. **Models Implemented**

The following classification models were implemented and compared:

Logistic Regression

Decision Tree

Random Forest

Support Vector Machine (SVM)

XGBoost

7. **Evaluation**

The models were evaluated using:

Accuracy Precision

Recall

F1-Score

Confusion Matrix

Model Performance Summary

Logistic Regression → ~94%

Decision Tree → ~92%

Random Forest → ~94%

SVM → ~92–93%

XGBoost → ~95–96% (Best Model)

Among all models, XGBoost achieved the highest accuracy with balanced precision and recall, making it the most reliable model for this dataset.

8. **Conclusion**

In this project, an end-to-end Credit Default Risk Prediction system was successfully developed using machine learning techniques. The complete data science workflow—from data preprocessing and EDA to model training and evaluation—was implemented.

Among all models, XGBoost delivered the best performance with an accuracy of approximately 95–96%, indicating strong predictive capability. This demonstrates that machine learning can effectively support financial institutions in predicting credit default risk and improving lending decisions.

9. **Business Impact**

This model can help financial institutions to:

Identify high-risk customers in advance

Reduce loan default rates

Improve credit approval decisions Minimize

financial losses

Ensure fair and consistent lending

10. **Future Improvements**

Although the current model achieved strong performance, it can be further improved in the future.

Hyperparameter tuning using GridSearchCV was explored; however, due to computational resource limitations, the execution was time-consuming and complete output could not be generated. With access to higher computational resources, extensive hyperparameter tuning can be performed to further improve model performance.

Additional future enhancements include:

Handling class imbalance using **SMOTE**

Adding **model explainability** using SHAP values

Evaluating performance using **ROC-AUC and Precision-Recall curves**

Deploying the model as a **web application** using Flask or FastAPI (Hyperparameter tuning was attempted, but full results could not be included due to computational limitations.)

11. **Final Note**

This project demonstrates the ability to:

Handle real-world structured data

Apply multiple machine learning algorithms

Evaluate and compare model performance

Build a complete prediction-ready machine learning pipeline